

Quiz (Habanero)

- Q1.** What is the discriminant of the quadratic equation $x^2 + 4x + 4 = 0$?
- (A) 0
(B) 4
(C) 8
(D) 12
(E) 16
- Q2.** The quadratic equation $x^2 - 7x + 12 = 0$ has how many solutions?
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 0
- Q3.** The quadratic equation $x^2 + 2x + 5 = 0$ has solutions of what type?
- (A) Real and rational
(B) Real and irrational
(C) Complex
(D) Imaginary
(E) No solutions
- Q4.** Solve for x in the equation $x^2 - 9x + 20 = 0$ using the quadratic formula. What is the value of x ?
- (A) 4 or 5
(B) 5 or 4
(C) -4 or -5
(D) -5 or -4
(E) 0 or 1
- Q5.** A water tank can hold 1200 gallons of water. Due to a leak, the tank is losing water at a rate of $2x^2 - 5x + 1$ gallons per hour. How many hours will it take for the tank to be empty, given the leak rate is modeled by the quadratic equation $2x^2 - 5x + 1 = 0$?
- (A) 2 hours
(B) 3 hours
(C) 4 hours
(D) 5 hours
(E) 1 hour
- Q6.** What is the discriminant of the quadratic equation $x^2 - 2x - 15 = 0$?
- (A) 49
(B) 64
(C) -49
(D) -64
(E) 1
- Q7.** A quadratic equation of the form $ax^2 + bx + c = 0$ has a discriminant of $b^2 - 4ac = 100$. What are the solutions of this equation?
- (A) Real and rational
(B) Real and irrational
(C) Complex
(D) Imaginary
(E) No solutions
- Q8.** The quadratic equation $x^2 + 5x + 6 = 0$ can be solved using the quadratic formula. What are the solutions to this equation?
- (A) -2 or -3
(B) -3 or -2
(C) 2 or 3
(D) 3 or 2
(E) 1 or 0

Open Ended Questions (FRQ)

- Q9.** In a certain ecosystem, the population of rabbits P after x years is modeled by the quadratic equation $P = 2x^2 + 5x + 10$. If the population of rabbits is 48, find the values of x . Show all work and explain your reasoning.
- Q10.** The height h of a projectile after x seconds is modeled by the quadratic equation $h = -x^2 + 4x + 10$. Determine the time it takes for the projectile to reach its maximum height and the value of that height. Use the quadratic formula to solve for x .

Chef's Tips

- Tip 1: Ensure all work is shown for free response questions.
- Tip 2: Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve quadratic equations.
- Tip 3: Check the discriminant $b^2 - 4ac$ to determine the type and number of solutions for each quadratic equation.